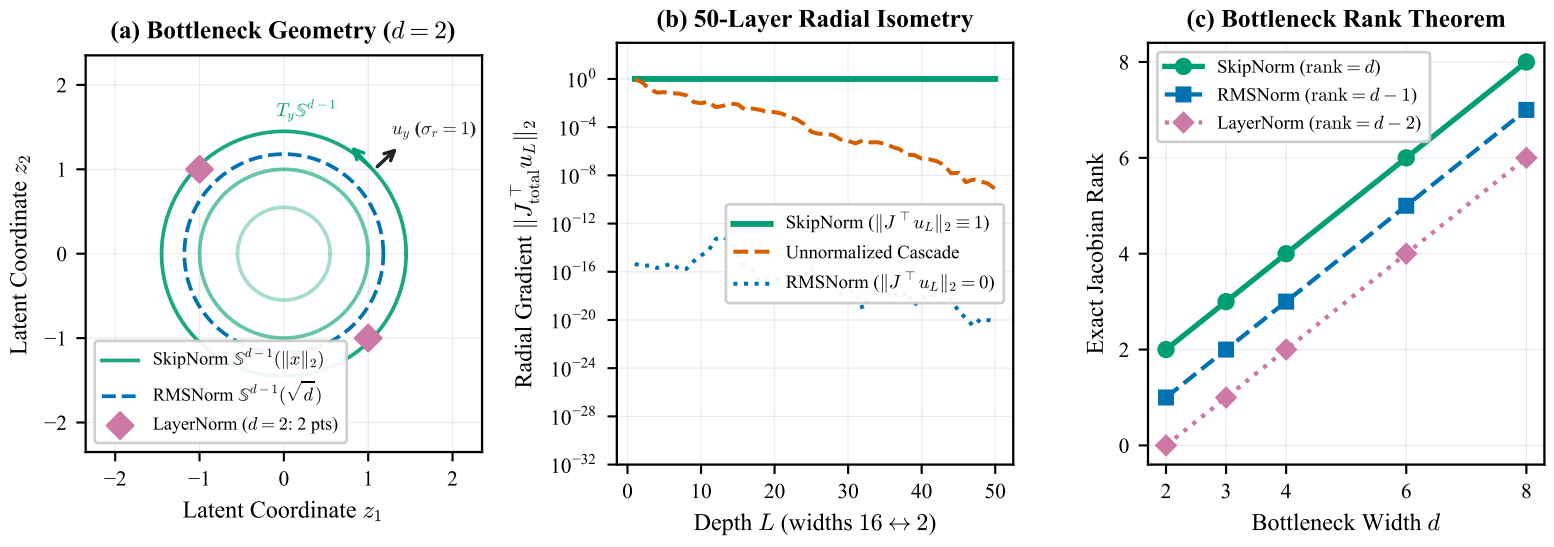

(a) Bottleneck Geometry ($d = 2$)
$T_y\mathbb{S}^{d-1}$
u_y ($\sigma_r = 1$)
SkipNorm $\mathbb{S}^{d-1}(\|x\|_2)$
RMSNorm $\mathbb{S}^{d-1}(\sqrt{d})$
LayerNorm ($d = 2$: 2 pts)
Latent Coordinate z_1
Latent Coordinate z_2
(b) 50-Layer Radial Isometry
SkipNorm ($\|J^\top u_L\|_2 \equiv 1$)
Unnormalized Cascade
RMSNorm ($\|J^\top u_L\|_2 = 0$)
Radial Gradient $\|J_{\text{total}}^\top u_L\|_2$
Depth L (widths $16 \leftrightarrow 2$)
(c) Bottleneck Rank Theorem
SkipNorm (rank $= d$)
RMSNorm (rank $= d - 1$)
LayerNorm (rank $= d - 2$)
Exact Jacobian Rank
Bottleneck Width d